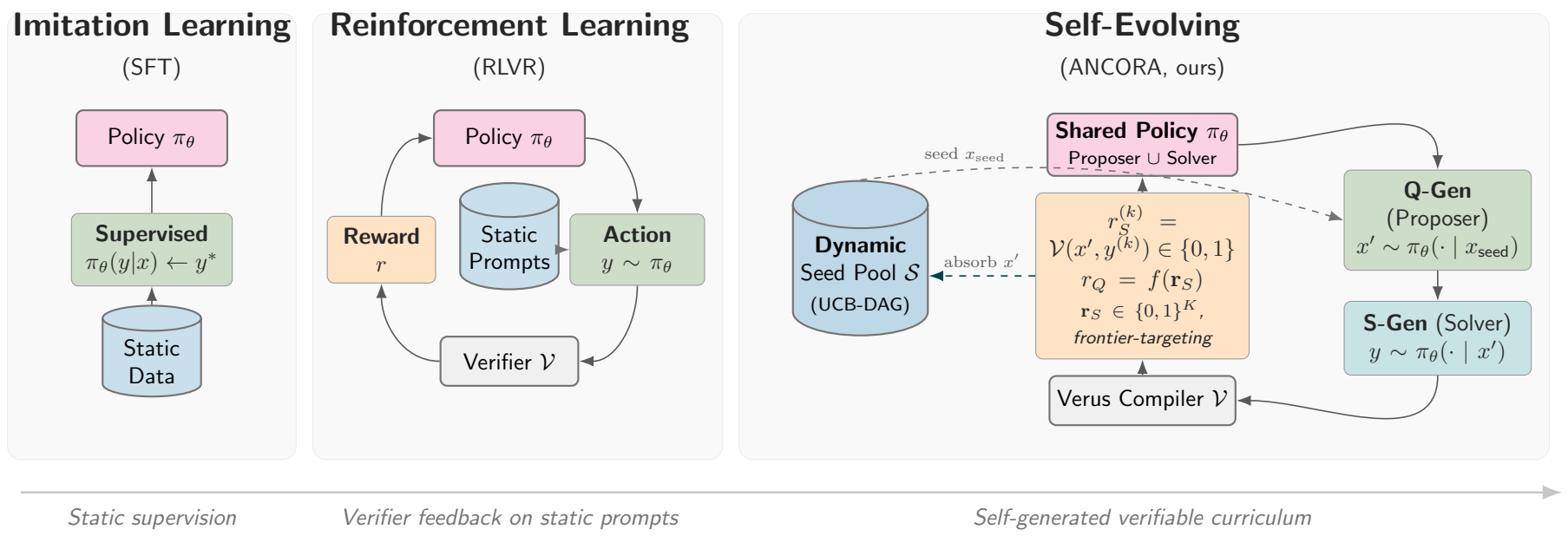

Imitation Learning
(SFT)
Policy π_θ
Supervised
$\pi_\theta(y|x) \leftarrow y^*$
Static Data
Reinforcement Learning
(RLVR)
Policy π_θ
Reward
r
Static Prompts
Action
$y \sim \pi_\theta$
Verifier $\mathcal{V}$
Self-Evolving
(ANCORA, ours)
Shared Policy π_θ
Proposer ∪ Solver
seed x_{seed}
Dynamic
Seed Pool $\mathcal{S}$
(UCB-DAG)
absorb x'
$r_S^{(k)} = \mathcal{V}(x', y^{(k)}) \in \{0,1\}$
$r_Q = f(\mathbf{r}_S)$
$\mathbf{r}_S \in \{0,1\}^K$,
frontier-targeting
Verus Compiler $\mathcal{V}$
Q-Gen
(Proposer)
$x' \sim \pi_\theta(\cdot \mid x_{\text{seed}})$
S-Gen (Solver)
$y \sim \pi_\theta(\cdot \mid x')$
Static supervision
Verifier feedback on static prompts
Self-generated verifiable curriculum